Micro Set 3. Negative Externalities — Model Answer Scheme

Question 1 [2 marks]

A negative externality in production occurs when the act of producing a good imposes costs on third parties not party to the transaction, such that the marginal social cost (MSC) of production exceeds the marginal private cost (MPC). *[1 mark.]*

Evidence: Extract 1.1 notes that "the carbon tax… makes polluters pay the social cost of their emissions," explicitly identifying that emissions impose a social cost on third parties (the global community affected by climate change) that producers do not bear privately. *[1 mark.]*

Examiner Note. The definition must specify the wedge between MPC and MSC. "Cost imposed on others" alone is too vague — it must use the marginal-cost framework.

Question 2 [2 marks]

The EV share of new registrations rose monotonically and sharply, from 0.2% in 2019 to 33.6% in 2024 — a roughly 168-fold increase. *[1 mark direction + magnitude.]* The acceleration was most pronounced from 2021 onward, with the share more than doubling each year from 2020 to 2022 (0.9% → 4.6% → 11.7%). *[1 mark for inflection.]*

Question 3 [4 marks]

Why S$5/t$CO_2$e produced only 5% emissions reduction.

A Pigouvian tax incentivises emissions reductions only up to the point where the marginal cost of abatement equals the tax. At S$5/t$CO_2$e, this threshold is far below the marginal abatement cost for most heavy industrial emitters — estimated at US$30–80 per tonne for chemicals and refining processes. Firms therefore prefer to pay the tax rather than invest in costly abatement, and emissions reduce only marginally. *[2 marks for MAC-vs-tax mechanism.]*

Extract 1.1 confirms that "abatement options for the chemicals and refining sectors require substantial capital investment with long lead times." These investments (heat-pump conversion, electrification of industrial furnaces, hydrogen-fuel substitution) are multi-year capital projects with uncertain returns. Firms rationally delay investment until the long-term tax trajectory is clear and material. *[1 mark for delay/option-value argument.]*

Additionally, the 80% emissions coverage means some emitters (those below 25 ktCO_2e/year threshold) are unaffected entirely, and the tax does not extend to consumption-side emissions (e.g. household energy). *[1 mark for coverage limitation.]*

Examiner Note. The marginal-abatement-cost-vs-tax framework is the key analytical move. Without it, the answer is descriptive ("too small a tax") rather than analytical ("below the MAC threshold").

Question 4 [4 marks]

VES rebate as externality internalisation.

The VES rebate effectively reduces the private cost of purchasing a low-emission vehicle by up to S$25000. Equivalently, it raises the relative private benefit of choosing a low-emission vehicle over a polluting alternative. *[1 mark for rebate mechanism.]*

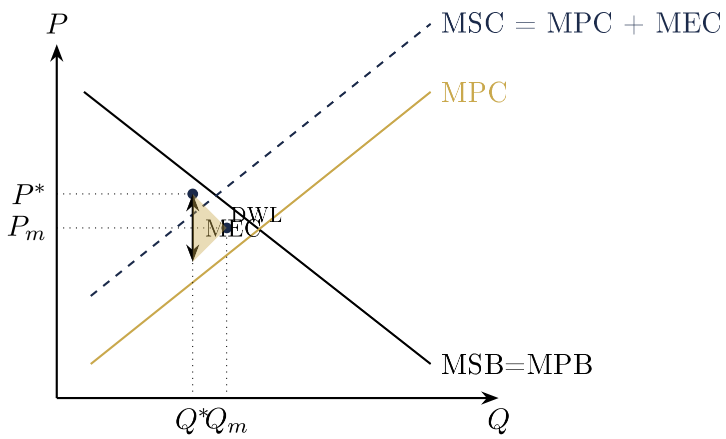
For a negative consumption externality (vehicle emissions impose health and environmental costs on the public not borne by the car buyer), MPB > MSB at every output. The rebate is a subsidy that internalises the externality *on the substitute* — it shifts consumer preferences toward goods that do not generate the externality, reducing the quantity of polluting vehicles consumed. *[2 marks for MPB/MSB framework.]*

In effect, the VES rebate shifts the relative demand toward clean vehicles. Extract 1.1 confirms the empirical magnitude: EV share rose from 0.2% (2019) to 33.6% (2024), demonstrating that the rebate has substantially altered consumer behaviour at the margin. *[1 mark for evidence anchor.]*

Examiner Note. The cleanest interpretation: the surcharge *taxes* polluting vehicles (raising MPC of consuming them) while the rebate *subsidises* clean vehicles. Both arms work together as a feebate. Mentioning both arms earns full marks.

Question 5 [6 marks]

Pigouvian tax to correct negative production externality.



In the diagram, MPC is the private supply curve reflecting the firm's marginal cost of production. MSC = MPC + MEC, where MEC is the marginal external cost (carbon emissions causing climate damage). The vertical distance between MSC and MPC represents the per-unit external cost. *[2 marks for diagram fully labelled with MEC wedge.]*

Without intervention, the free market equates MPB and MPC, producing Q_m at price P_m. This output exceeds the socially optimal level Q^*, where MSB = MSC. The over-allocation $|Q_m - Q^*|$ generates a deadweight loss (shaded), equal to the triangle between MSB and MSC over the range $[Q^*, Q_m]$. *[2 marks for Q_m vs Q^* + DWL.]*

A Pigouvian tax set equal to MEC at Q^* shifts MPC upward to coincide with MSC. Firms now internalise the external cost when making production decisions. The new equilibrium output falls to Q^*, the socially optimal level. The deadweight loss is eliminated, and the tax revenue collected (rectangle area equal to MEC $\times Q^*$) can be used to compensate the harmed parties or fund further abatement. *[2 marks for tax-mechanism + welfare restoration.]*

Examiner Note. The fully labelled diagram is mandatory: axes (P, Q), all three curves (MPB, MPC, MSC), both equilibrium points (Q_m, Q^*), the MEC wedge clearly marked, and the DWL triangle shaded. Missing any element costs marks.

Question 6 [6 marks]

VES effectiveness vs carbon tax effectiveness.

Reason 1: Magnitude relative to private cost. The VES rebate of S$25000 is approximately 20–30% of the typical Singapore car price (S$100,000–150,000 for the cheaper bands after COE). This is a material proportion of the private cost, large enough to shift consumer choices substantially. In contrast, the S$5/t$CO_2$e carbon tax (2019–2023) was approximately 0.5–2% of typical industrial production cost, far too small to drive substantial behaviour change. The VES's price signal is therefore an order of magnitude more powerful relative to the underlying decision. *[2 marks: magnitude comparison with values.]*

Reason 2: Visibility and decision-timeframe. Consumer car purchases are individual, salient, one-time decisions where the price signal is fully visible at the point of purchase. Firms making industrial decisions face a multi-year capital horizon: emissions reductions require heat-pump installations, hydrogen substitution, or electrification, with payback periods of 5–15 years. The mismatch between the tax horizon (annual) and the investment horizon (decade-scale) creates option-value delay. The carbon tax was raised only in 2024, so firms rationally waited rather than committing capital. *[2 marks: visibility + investment horizon.]*

Reason 3: Direct substitutability of alternatives. EVs have become a directly available substitute for petrol cars, supported by Singapore's expanding charging network. The consumer choice is binary and immediate. In contrast, industrial emitters face complex substitution choices involving multiple technology paths (e.g. blue vs green hydrogen, electrification vs CCS), each with substantial uncertainty. *[2 marks: substitute availability.]*

Examiner Note. Strong 6-mark policy-comparison answers identify three distinct dimensions, with quantitative anchors where possible. "EVs are cheaper now" alone earns 2 marks at most.

End of Set 3 — Model Answers